Feier Lyu (Fay Lyu)

Optimization theory and formal verification

Academic profile & contact | GitHub | ORCID

Education

BSc Mathematics, University of Edinburgh
Exchange student at Cornell University, 2025-2026
Expected graduation: 2027

Research Interests

Optimization theory and formal verification. Current projects formalize submodular greedy methods and projected gradient descent in Isabelle/HOL, and study second-order geometry in network localization.

Publications

Feier Lyu. "First-Order Methods for Smooth Convex Optimization in Isabelle/HOL."

Archive of Formal Proofs · July 2026

An Isabelle/HOL formalization of smooth convex first-order optimization.

AFP entry · Proof document · Project note · GitHub

Feier Lyu. "Greedy Algorithms for Cardinality-Constrained Submodular Maximization."

Archive of Formal Proofs · May 2026

With feedback and guidance from Wenda Li.

AFP entry · Proof document · DOI

Research Experience

Research Collaboration with Shoham Sabach and Owen Li

2026-Present

Ongoing collaboration on network structure and second-order geometry in network localization. Owen Li and I both contribute theoretical derivations and numerical experiments, studying how graph structure and node geometry affect second-order behavior at differentiable stationary points. Technical results are not yet public.

Patents

Anti-Segregation Spreading Method and Device (descriptive English translation)

Role: Co-inventor. Chinese invention patent application, Application No. 202411275605.1, pending.

Related utility model: Anti-Segregation Spreading Device (descriptive English translation); role: Co-inventor; Patent No. ZL202422234130.3, granted July 4, 2025.

Designed a mechanical system to improve material homogeneity in industrial spreading processes.

Industry Experience

Data Engineering Intern

Liblib AI (Beijing Qidian Xingyu Technology Co., Ltd.)
Jun 2025 - Jul 2025

Wrote SQL in Alibaba Cloud DataWorks to answer data requests from the operations team, including multi-source joins, filtering, aggregation, and result validation.